

Data-volume calibration for resolving the three-mode slow relaxation band

Frozen synthetic-covariance audit

9 September 2026

Abstract

We calibrate how many independent trajectories are required to separate two nearby real relaxation rates using the empirical cross-lag covariance of the controlled three-mode data. The complete frozen grid comprises 336 cells and 168,000 nonlinear-fit replicates. The target spacing $\Delta = 0.20$ fails for both driven and equilibrium covariance at every tested stream count through $N = 2048$, for both fit windows and both amplitude constraints. The predeclared Part-2 gate is therefore closed and no additional physical trajectory simulation is launched. The result is a quantitative non-resolution bound, not evidence that the physical slow band is absent.

1 Question and frozen protocol

The synthetic truth is

$$C(t) = 0.6e^{-0.90t} + 0.4e^{(-0.90-\Delta)t},$$

with $\Delta \in \{0.05, 0.10, 0.15, 0.20, 0.30, 0.40, 0.60\}$ and $N \in \{64, 128, 256, 512, 1024, 2048\}$. Noise preserves the full empirical cross-lag covariance of the saved I_2 correlation curves and scales as $\sqrt{64/N}$. We use 500 deterministic replicates per case- N - Δ -window-constraint cell. Real sums of one to four exponentials are selected sequentially by $\Delta\text{AIC} \leq -10$. A cell resolves the two rates only if at least 90% of successful fits select $m \geq 2$ and both median matched-rate errors are at most $\max(0.05, \Delta/4)$. The fixed window is $[0.20, 4.00]$; the extended window follows the frozen contiguous $C/\text{SE} \geq 3$ rule. No rule was changed after output existed.

2 Primary result: the Part-2 gate is closed

None of the 48 cells at the target spacing $\Delta = 0.20$ passes. Consequently there is no common tested $N \leq 2048$, window, and amplitude rule that resolves the target in both bath cases. Under the frozen protocol, Part 2 is forbidden.

Case	Window	amplitudes	smallest resolved at $N = 2048$	resolved cells
Driven (2,8)	fixed	signed	0.60	1
Driven (2,8)	fixed	nonnegative	0.40	2
Driven (2,8)	extended	signed	0.60	1
Driven (2,8)	extended	nonnegative	0.60	1
Equilibrium (5,5)	fixed	signed	0.60	1
Equilibrium (5,5)	fixed	nonnegative	0.60	1
Equilibrium (5,5)	extended	signed	0.60	1
Equilibrium (5,5)	extended	nonnegative	0.60	1

Table 1: Observed resolution at the largest tested stream count.

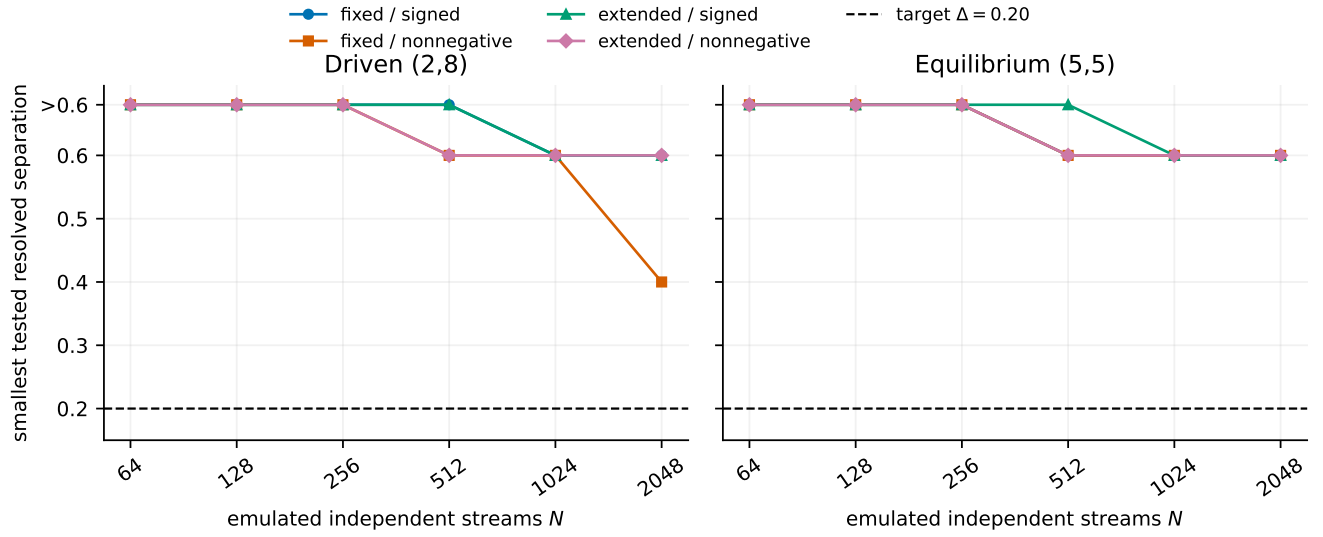


Figure 1: Smallest tested separation satisfying the complete resolution rule. Values marked > 0.6 mean no tested separation passed. The dashed line is the target $\Delta = 0.20$.

3 Why increasing N did not resolve $\Delta = 0.20$

The probability of selecting multiple exponentials approaches one, but the matched rate errors remain well above the 0.05 tolerance. Simultaneously, the original numerical-identifiability fraction generally decreases. Model-order selection alone is therefore not evidence that the two physical rates have been recovered.

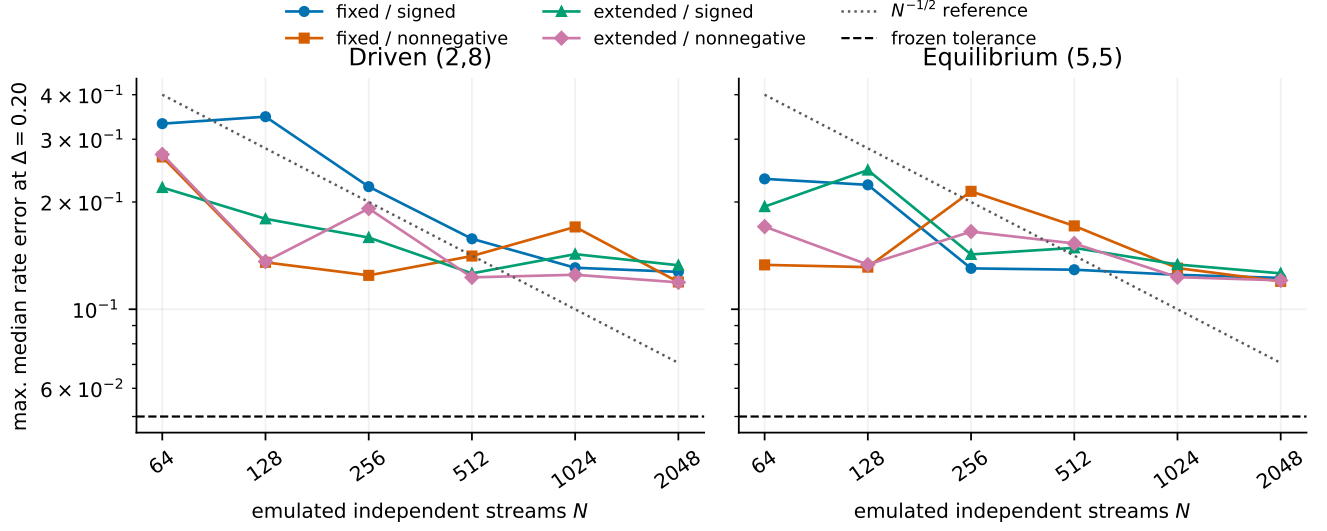


Figure 2: Maximum of the two median matched-rate errors at $\Delta = 0.20$. The dotted $N^{-1/2}$ curve is a slope reference, not a fit; the dashed line is the frozen tolerance.

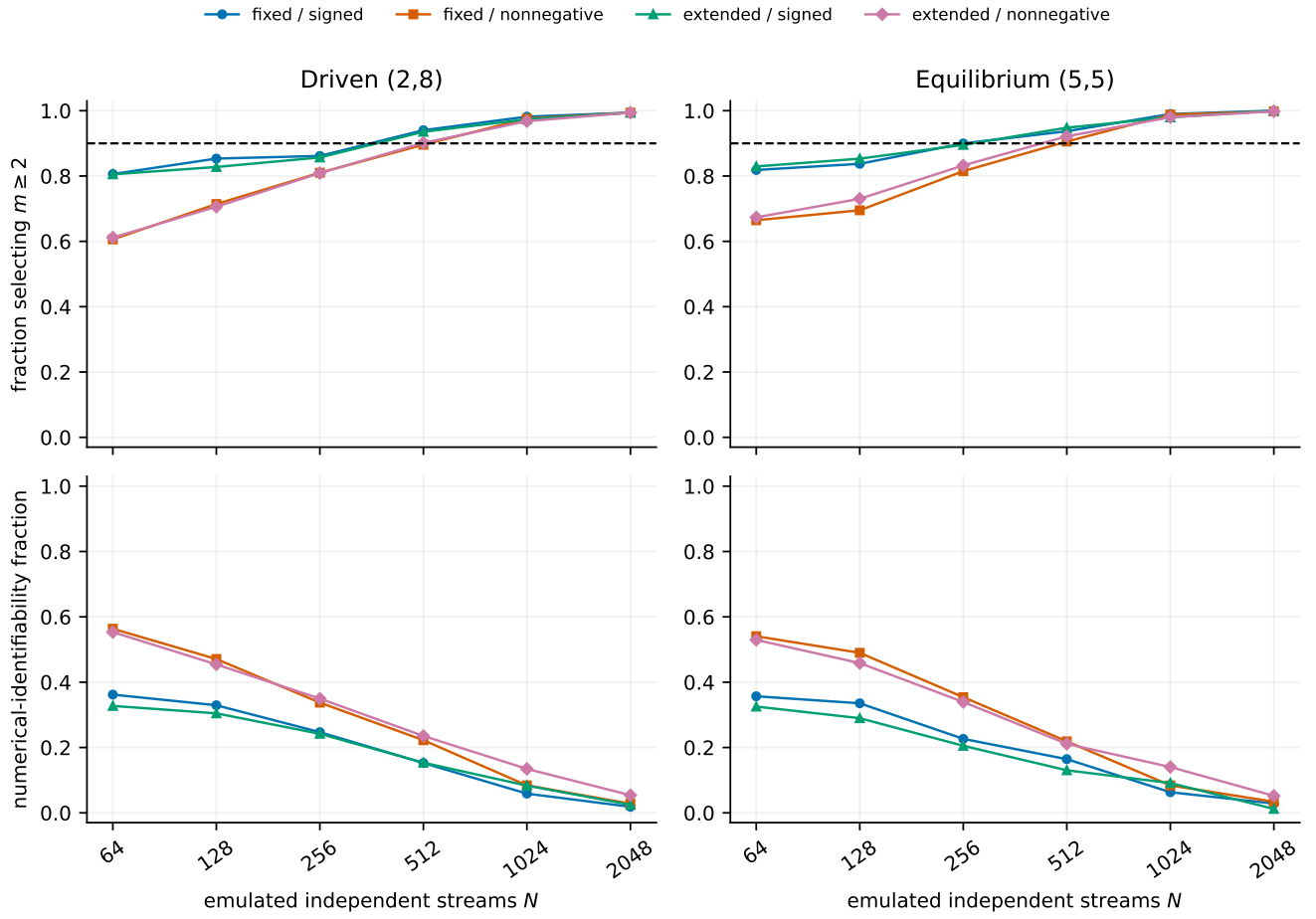


Figure 3: At $\Delta = 0.20$, selection of $m \geq 2$ becomes frequent while numerical identifiability deteriorates.

4 All target-spacing summaries

Case	Window	amplitudes	N	$P(m \geq 2)$	error slow	error fast	num.-ID	pass
Driven (2,8)	extended	nonnegative	64	0.612	0.272	0.125	0.553	no
Driven (2,8)	extended	nonnegative	128	0.706	0.136	0.126	0.454	no
Driven (2,8)	extended	nonnegative	256	0.809	0.192	0.125	0.350	no
Driven (2,8)	extended	nonnegative	512	0.901	0.076	0.123	0.235	no
Driven (2,8)	extended	nonnegative	1024	0.968	0.072	0.125	0.135	no
Driven (2,8)	extended	nonnegative	2048	0.994	0.065	0.119	0.054	no
Driven (2,8)	extended	signed	64	0.805	0.186	0.220	0.327	no
Driven (2,8)	extended	signed	128	0.828	0.148	0.179	0.305	no
Driven (2,8)	extended	signed	256	0.857	0.159	0.159	0.242	no
Driven (2,8)	extended	signed	512	0.935	0.088	0.126	0.154	no
Driven (2,8)	extended	signed	1024	0.974	0.070	0.143	0.083	no
Driven (2,8)	extended	signed	2048	0.994	0.066	0.133	0.025	no
Driven (2,8)	fixed	nonnegative	64	0.606	0.268	0.128	0.564	no
Driven (2,8)	fixed	nonnegative	128	0.714	0.135	0.128	0.471	no
Driven (2,8)	fixed	nonnegative	256	0.810	0.121	0.124	0.337	no
Driven (2,8)	fixed	nonnegative	512	0.896	0.141	0.123	0.222	no
Driven (2,8)	fixed	nonnegative	1024	0.976	0.170	0.123	0.084	no
Driven (2,8)	fixed	nonnegative	2048	0.994	0.085	0.119	0.028	no
Driven (2,8)	fixed	signed	64	0.806	0.144	0.332	0.362	no
Driven (2,8)	fixed	signed	128	0.853	0.101	0.347	0.329	no
Driven (2,8)	fixed	signed	256	0.861	0.096	0.221	0.247	no
Driven (2,8)	fixed	signed	512	0.940	0.075	0.158	0.153	no
Driven (2,8)	fixed	signed	1024	0.982	0.072	0.131	0.059	no
Driven (2,8)	fixed	signed	2048	0.994	0.067	0.127	0.019	no
Equilibrium (5,5)	extended	nonnegative	64	0.673	0.171	0.132	0.529	no
Equilibrium (5,5)	extended	nonnegative	128	0.730	0.083	0.133	0.458	no
Equilibrium (5,5)	extended	nonnegative	256	0.832	0.165	0.127	0.339	no
Equilibrium (5,5)	extended	nonnegative	512	0.922	0.153	0.126	0.211	no
Equilibrium (5,5)	extended	nonnegative	1024	0.980	0.071	0.123	0.140	no
Equilibrium (5,5)	extended	nonnegative	2048	0.998	0.067	0.121	0.052	no
Equilibrium (5,5)	extended	signed	64	0.829	0.194	0.162	0.325	no
Equilibrium (5,5)	extended	signed	128	0.853	0.089	0.246	0.290	no
Equilibrium (5,5)	extended	signed	256	0.895	0.129	0.143	0.205	no
Equilibrium (5,5)	extended	signed	512	0.948	0.149	0.126	0.130	no
Equilibrium (5,5)	extended	signed	1024	0.980	0.069	0.134	0.091	no
Equilibrium (5,5)	extended	signed	2048	1.000	0.065	0.126	0.012	no
Equilibrium (5,5)	fixed	nonnegative	64	0.665	0.123	0.133	0.541	no
Equilibrium (5,5)	fixed	nonnegative	128	0.695	0.078	0.131	0.490	no
Equilibrium (5,5)	fixed	nonnegative	256	0.815	0.214	0.126	0.354	no
Equilibrium (5,5)	fixed	nonnegative	512	0.906	0.171	0.124	0.219	no
Equilibrium (5,5)	fixed	nonnegative	1024	0.988	0.130	0.123	0.084	no
Equilibrium (5,5)	fixed	nonnegative	2048	0.998	0.081	0.120	0.034	no
Equilibrium (5,5)	fixed	signed	64	0.819	0.118	0.232	0.357	no
Equilibrium (5,5)	fixed	signed	128	0.837	0.077	0.223	0.335	no
Equilibrium (5,5)	fixed	signed	256	0.900	0.091	0.130	0.226	no
Equilibrium (5,5)	fixed	signed	512	0.937	0.074	0.129	0.164	no
Equilibrium (5,5)	fixed	signed	1024	0.990	0.071	0.125	0.063	no
Equilibrium (5,5)	fixed	signed	2048	1.000	0.064	0.122	0.029	no

5 Scaling, saturation, and descriptive extrapolation

For each case and rule, a descriptive power law was fit to the maximum median error over $N \geq 256$. Seven of eight curves meet the frozen saturation rule: the target remains unresolved at $N = 2048$ and the error decreases by less than 10% from $N = 1024$ to 2048. Exponents are much shallower than the independent-stream benchmark $-1/2$. Extrapolated N values do not include the separate 90% model-selection requirement and cannot authorize production.

Case	Window	amplitudes	exponent	implied N	reduction	saturated
Driven (2,8)	fixed	signed	-0.265	51659	2.6%	yes
Driven (2,8)	fixed	nonnegative	0.009	∞	29.9%	no
Driven (2,8)	extended	signed	-0.060	22365519689	6.9%	yes
Driven (2,8)	extended	nonnegative	-0.204	100666	4.8%	yes
Equilibrium (5,5)	fixed	signed	-0.031	4744167290403765	2.0%	yes
Equilibrium (5,5)	fixed	nonnegative	-0.291	35137	8.2%	yes
Equilibrium (5,5)	extended	signed	-0.068	1893455711	5.5%	yes
Equilibrium (5,5)	extended	nonnegative	-0.168	323382	2.1%	yes

6 Audit and reproducibility

The integrity audit is **PASS**. It reconstructs all 336 summaries from the 168,000 raw rows, verifies exactly 500 replicate indices in every cell, reproduces the Part-2 gate, and verifies the source and both input hashes.

```
python3 experiments/slow_band_resolution_scaling_2026-09-07/run_scaling.py
--output experiments/slow_band_resolution_scaling_2026-09-07/part1_analysis
--replicates 500 --workers 8
```

Master seed: 2026091201. Runtime: 27.55 hours. Analysis source SHA-256: b2ad9a90fc9ecfd2799c9b778dff3d7194c28faca224da70f20f5db022dce1d4.

- Driven input SHA-256: 9a878a637bd7475ab0f05ea0b7991276f4de1fd53f152c274530b7f99a984a47.
- Equilibrium input SHA-256: f7f97e860d75414b3999d656cf585dea9d402fafe7cc1a99a22321e9963c3d45.
- Previous fitting pipeline SHA-256: ada94924e17f762519a973b28712a3af57ab6f9d1987183a2459c830587d94cb.

All raw replicates, cell summaries, task seeds, resolution curves, extrapolation diagnostics, hashes, and the frozen protocol accompany this report.

7 Claim boundary

The result supports only the following statement: under the empirical covariance and frozen fitting protocol, a separation of 0.20 is not recoverable with up to 2048 emulated independent streams. It does not establish that the physical slow band is absent, and it does not justify quoting a unique spectral gap. No real-data Part-2 simulation was run because its predeclared gate failed.